



GCI

SQL and Database

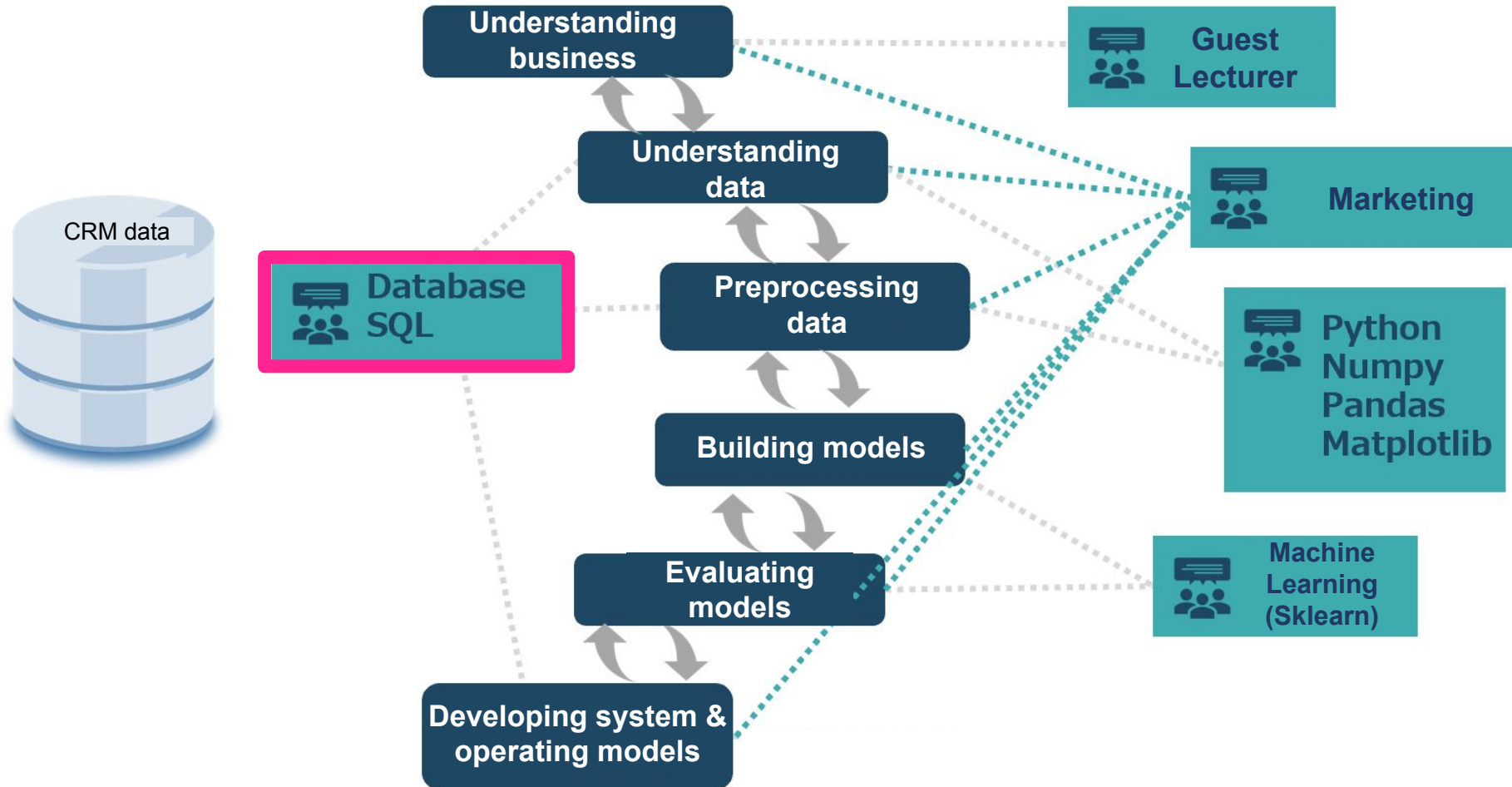


松尾・岩澤研究室

MATSUO-IWASAWA LAB UTOKYO

- Introduction to Database
- SQL
- Database and Data Science
- SQL Exercises

(From Week 1) Connection between GCI & real-life data science project

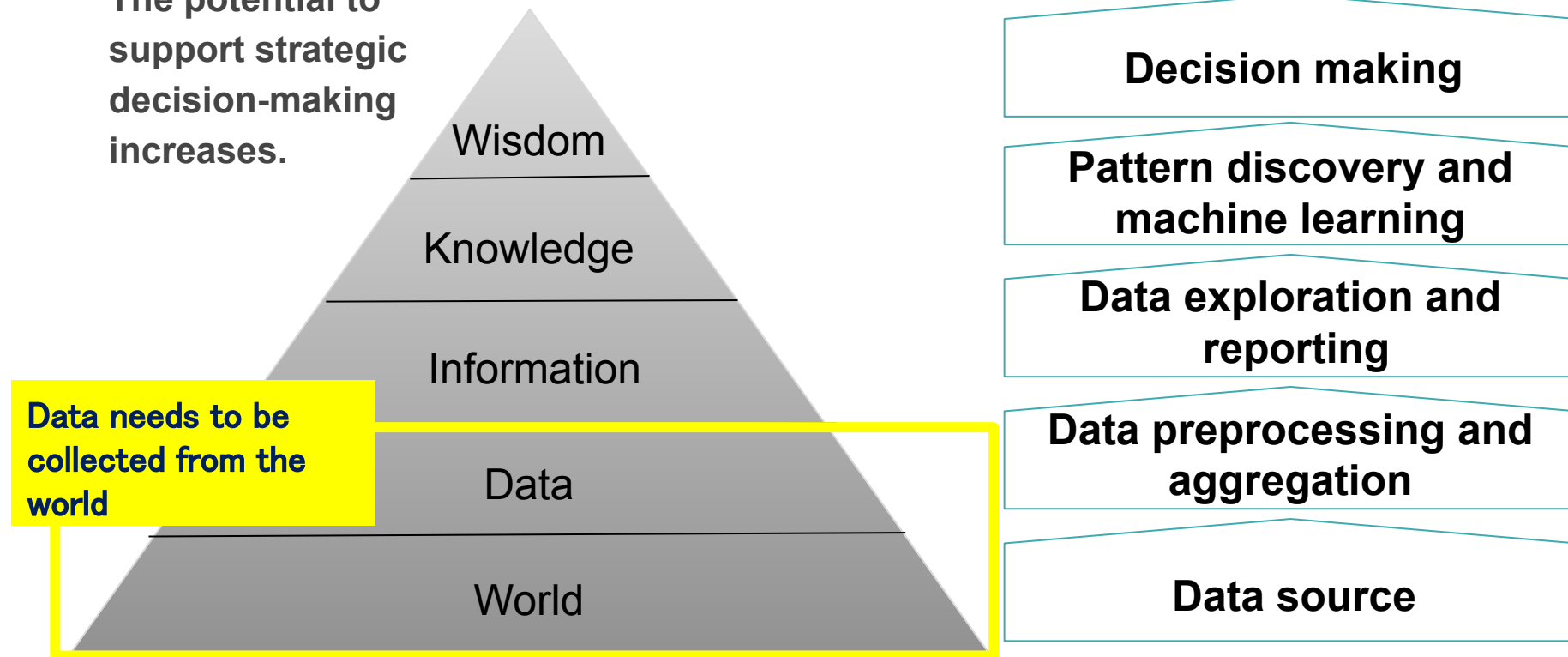


(From Week 1) Data analysis in decision-making



It is important to accumulate data and elevate it to wisdom to be used for decision-making

The potential to support strategic decision-making increases.



Reference: "Data Science" Newton Press

Examples of Data

- Transaction (deposit, withdrawal) data

Date	CustomerID	LocationNo.	Update	Total
2019/5/27 10:00:00.00	20001	20	2019/5/27/ 10:00:00.00	20000
2019/5/27 10:10:00.00	20010	25	2019/5/27/ 10:00:00.00	-20000
2019/5/27 10:11:00.00	20001	20	2019/5/27/ 10:00:00.00	36000

CustomerID=20001 deposited
¥20,000 on 2019/5/27 10AM

- Customer attribute data

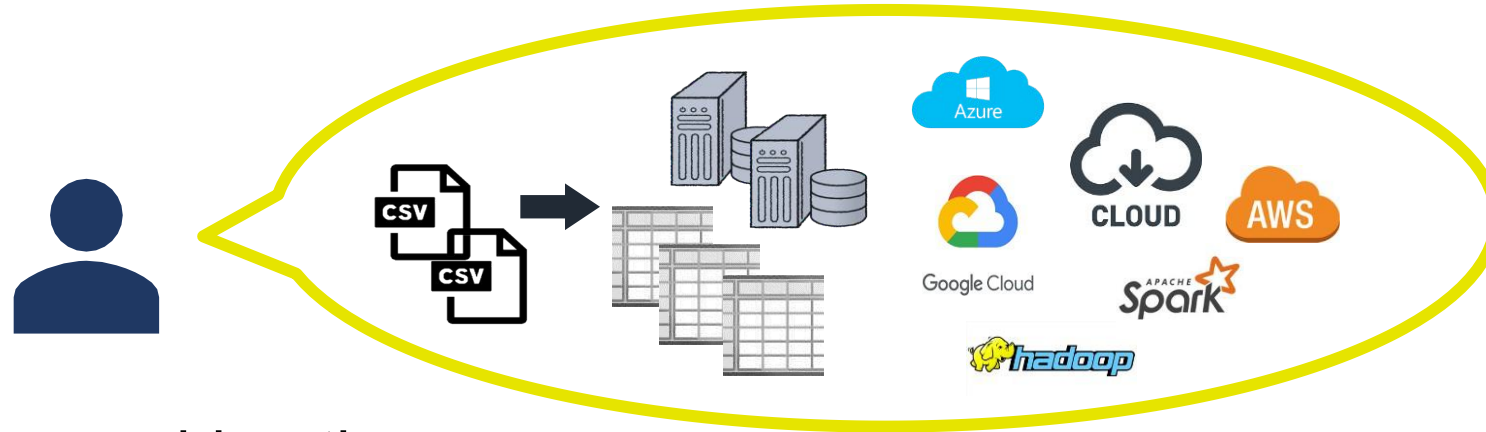
CustomerID	Gender	Occupancy	Update	Area
20001	F	100	2019/4/27/ 10:00:00.00	60
20010	M	13	2019/5/27/ 10:00:00.00	1344
20011	M	20	2019/5/27/ 10:00:00.00	491

Attributes of
CustomerID=20001

Introduction to Database

What is Database Design?

- Defining data processing and analysis strategies by
 - Aligning with objectives and business requirements
 - Designing data infrastructure and constructing databases

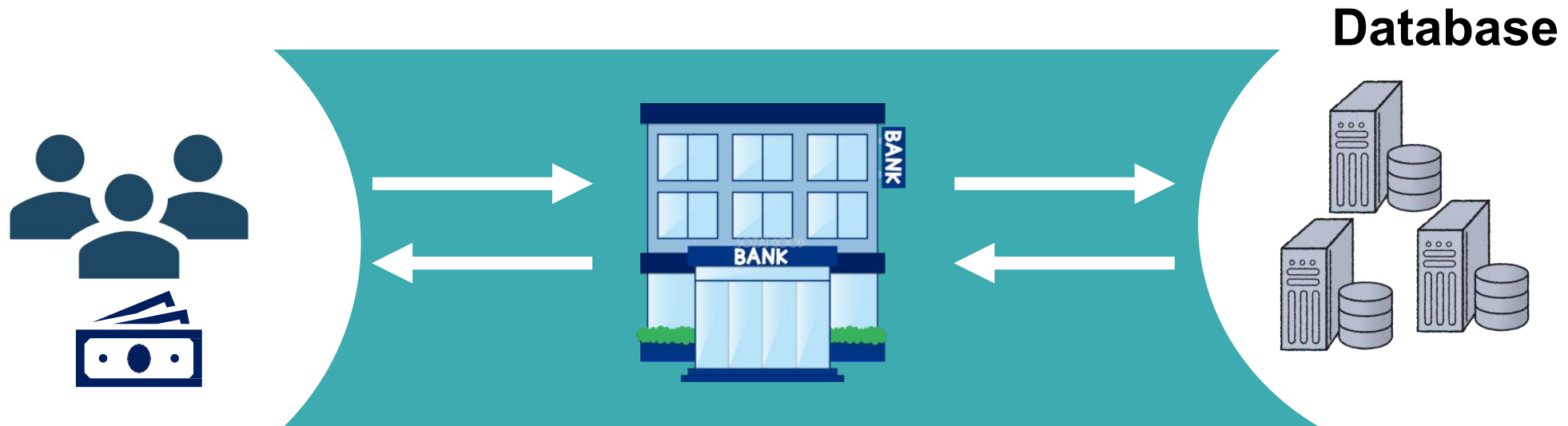


- Key considerations
 - Architecture to process data efficiently, optimized table design to prevent data duplication, knowledge about domain expertise and cloud services, etc.
- Keywords: database design, data modeling, database architect, database administrator, ER diagram, normalization, ETL

What is a Database?



- Databases are essential across various industries and sectors
- Examples
 - **Financial services:** ATM transactions, stock and forex trading
 - **Retail:** Purchases in stores (POS data: Point of Sales)
 - **E-Commerce:** Online shopping transactions (Amazon, etc.)
 - **Reservation system:** Booking flights, trains, and event tickets



Benefits of Using Databases



Data Consistency

- Integrity and eliminate duplications
- Referential integrity

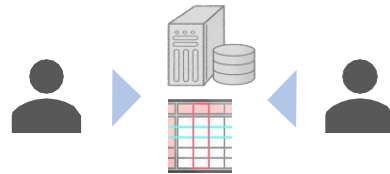
Data Durability

- Backup and recovery



Data Access

- Concurrency Control



Data Independence

- Separation from system processing

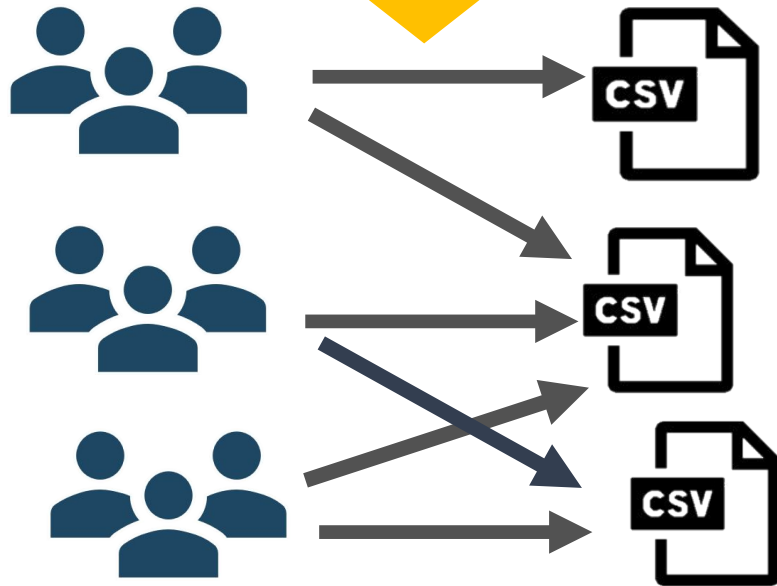
Enhanced Security

- Access controls and permissions
- Data Encryption

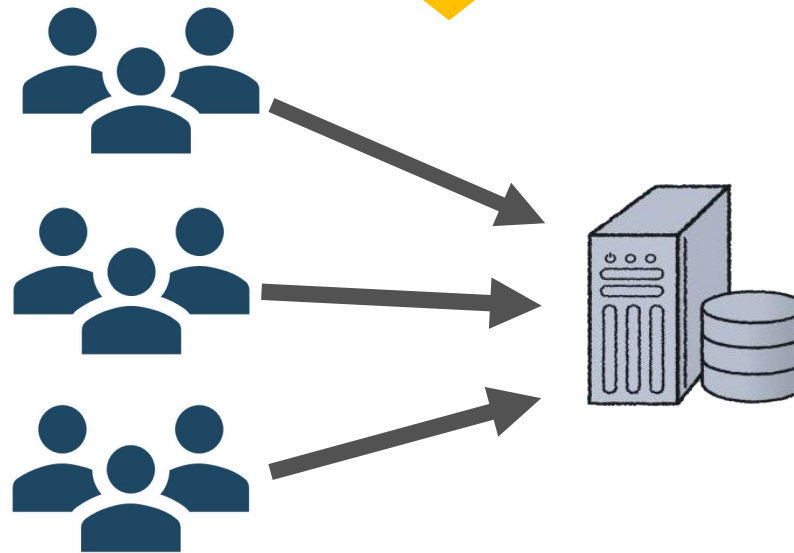
Benefits of Using Databases: Data Consistency

- Users can access data without having to think about where the data is stored
- Data is managed centrally = ensures data integrity

Managing data as individual files



Managing data as database



Types of Databases



Relational Database

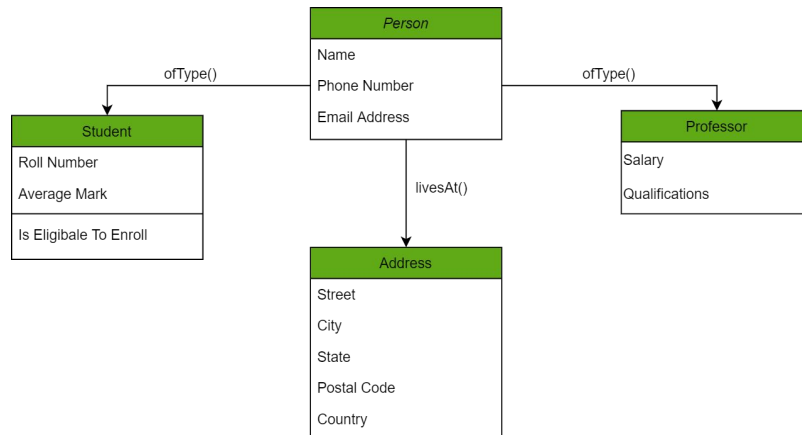
Roll no.	Student Name	Marks Awarded
1	Raman Tripathi	86
2	Rajan Govindan	94
3	Mahesh Nandalal	94

Key=94

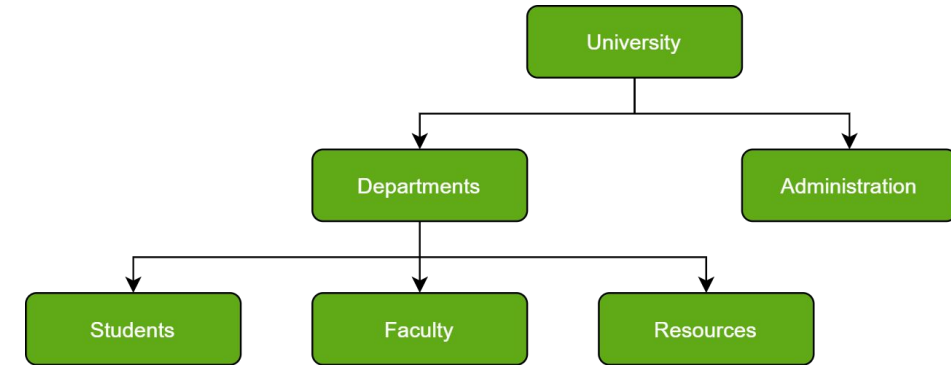
Section	Student Name	Marks Awarded	Rank
A	Raman Tripathi	86	43
B	Rajan Govindan	94	17
C	Mahesh Nandalal	94	16

Marks Awarded	Student Name	Rank	Scholarship
94	Rajan Govindan	17	Yes
94	Mahesh Nandalal	16	Yes

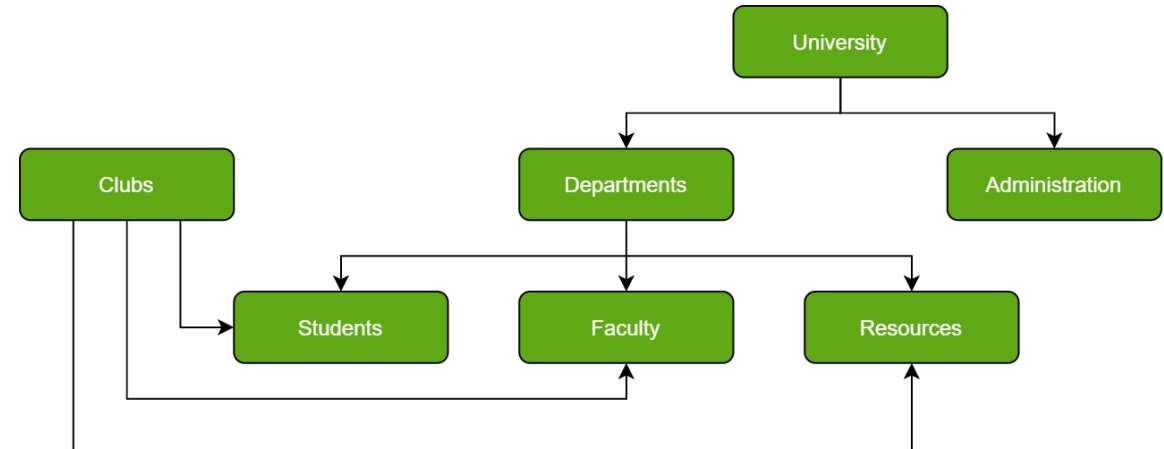
Object-oriented Database



Hierarchical Database



Network Database

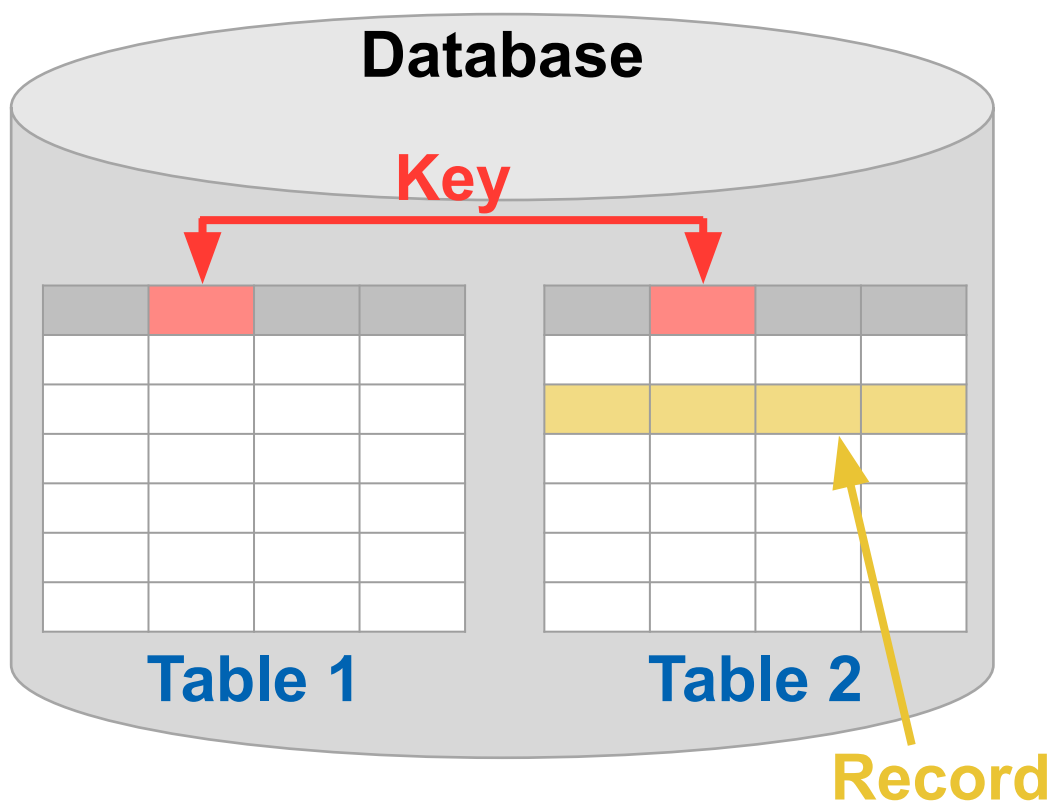


Other database examples XML database, NoSQL database, etc.

Reference: <https://www.geeksforgeeks.org/types-of-databases/>

Relational Database

- Organized as collections of two-dimensional **tables**
- Tables are connected through **keys** (unique identifier of each **record**)



1. Consumer Info Table

ID	birth_year	city	name
1	1989	Tokyo	Bob
2	1990	Osaka	Alice
3	1987	Kyoto	Akiko

2. Purchase History Table

Date	ID	Item	Price
2018-04-01	1	Notebook	108
2018-05-01	1	Computer	60000
2018-05-04	2	Bag	4000

Database Management System (DBMS)

- Software that provides functionalities to create, manage, and interact with databases
- Facilitates data storage, retrieval, update, and administration
- Types of DBMS:
 - Relational DBMS
 - Designed for structured table data
 - Non-relational DBMS
 - Flexible, designed for unstructured or sem-structured data



Reference:

<https://www.indiamart.com/global-standards-for/oracle-sql-mysql.html>

<https://www.decipherzone.com/blog-detail/nosql-databases>

SQL

What is SQL?



- SQL: Structured Query Language
- Programming language to manage relational databases
- Enables large-scale data processing (faster than Pandas in some cases)
- Popular SQL Services
 - Microsoft SQL Server
 - MySQL
 - PostgreSQL
 - SQLite

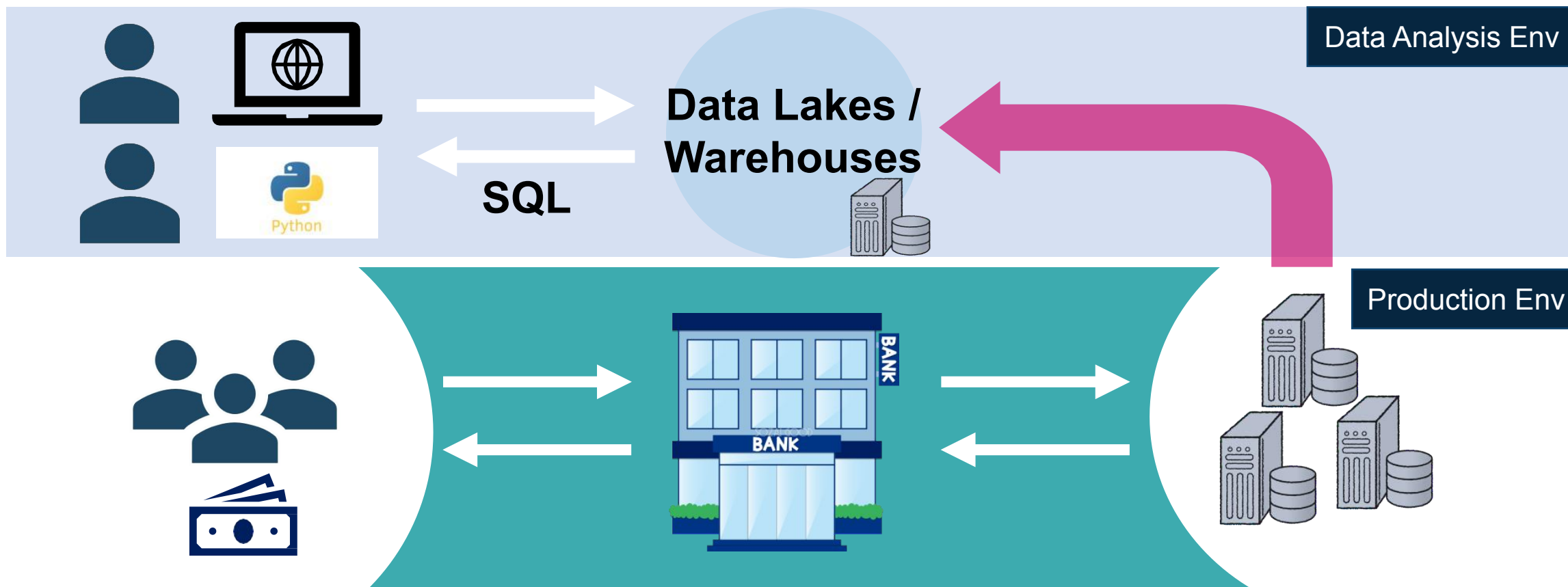


Reference: <https://www.indiamart.com/global-standards-for/oracle-sql-mysql.html>

Database and Data Science

Relationship Between Data Science and Databases

- Create data storage (data lakes, data warehouses), then preprocess and analyze the data using SQL or Python



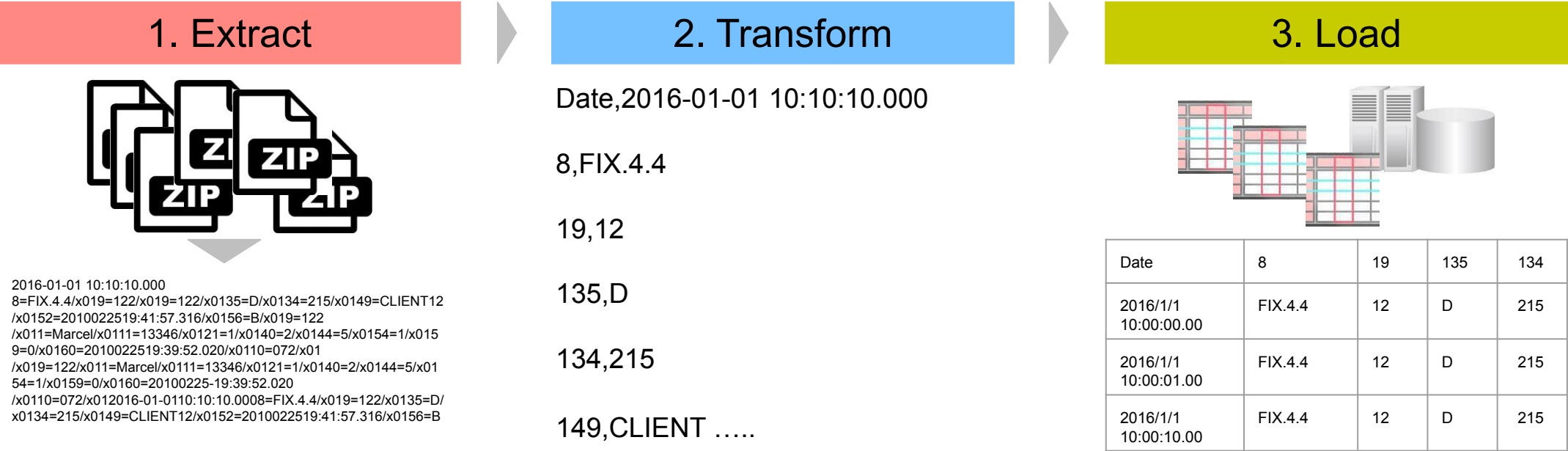
(From Week 1) Data could be disorganized

- Semi-structured financial data, terabyte-level per month, with over 2 billion rows
- 80-90% of data analysis process is spent on data processing

2016-01-01 10:10:10.000 8=FIX.4.4/x019=122/x019=122/x0135=D/x0134=215/x0149=CLIENT12/x0152=2010022519:41:57.316/x0156=B/x019=122/x011=Marcel/x0111=13346/x0121=1/x0140=2/x0144=5/x0154=1/x0159=0/x0160=2010022519:39:52.020/x0110=072/x019=122/x011=Marcel/x0111=13346/x0121=1/x0140=2/x0144=5/x0154=1/x0159=0/x0160=20100225-19:39:52.020/x0110=072/x012016-01-0110:10:10.0008=FIX.4.4/x019=122/x0135=D/x0134=215/x0149=CLIENT12/x0152=2010022519:41:57.316/x0156=B/x011=Marcel/x0111=13346/x0121=1/x0140=2/x0144=5/x0154=1/x0159=0/x0160=2010022519:39:52.020/x0110=072/x01/x019=122/x011=Marcel/x0111=13346/x0121=1/x0140=2/x0144=5/x0154=1/x0159=0/x0160=20100225-19:39:52.020/x0110=072/x01 2016-01-01 10:10:10.000 8=FIX.4.4/x019=122/x0135=D/x0134=215/x0149=CLIENT12/x0152=20100225-19:41:57.316/x0156=B/x011=Marcel/x0111=13346/x0121=1/x0140=2/x0144=5/x0154=1/x0159=0/x0160=20100225-19:39:52.020/x0110=072/x01/x019=122/x011=Marcel/x0111=13346/x0121=1/x0140=2/x0144=5/x0154=1/x0159=0/x0160=20100225-19:39:52.020/x0110=072/x01 2016-01-01 10:10:10.000 8=FIX.4.4/x019=122/x0135=D/x0134=215/x0149=CLIENT12/x0152=20100225-19:41:57.316/x0156=B/x011=Marcel/x0111=13346/x0121=1/x0140=2/x0144=5/x0154=1/x0159=0/x0160=20100225-19:39:52.020/x0110=072/x018=FIX.4.4/x019=122/x0135=D/x0134=215/x0149=CLIENT12/x0152=20100225-19:41:57.316/x0156=B/x011=Marcel/x0111=13346/x0121=1/x0140=2/x0144=5/x0154=1/x0159=0/x0160=20100225-19:39:52.020/x0110=072/x018=FIX.4.4/x019=122/x0135=D/x0134=215/x0149=CLIENT12/x0152=20100225-19:41:57.316/x0156=B/x011=Marcel/x0111=13346/x0121=1/x0140=2/x0144=5/x0154=1/x0159=0/x0160=20100225-19:39:52.020/x0110=072/x018=FIX.4.4/x019=122/x0135=D/x0134=215/x0149=CLIENT12/x0152=20100225-19:41:57.316/x0156=B/x011=Marcel/x0111=13346/x0121=1/x0140=2/x0144=5/x0154=1/x0159=0/x0160=20100225-19:39:52.020/x0110=072/x01

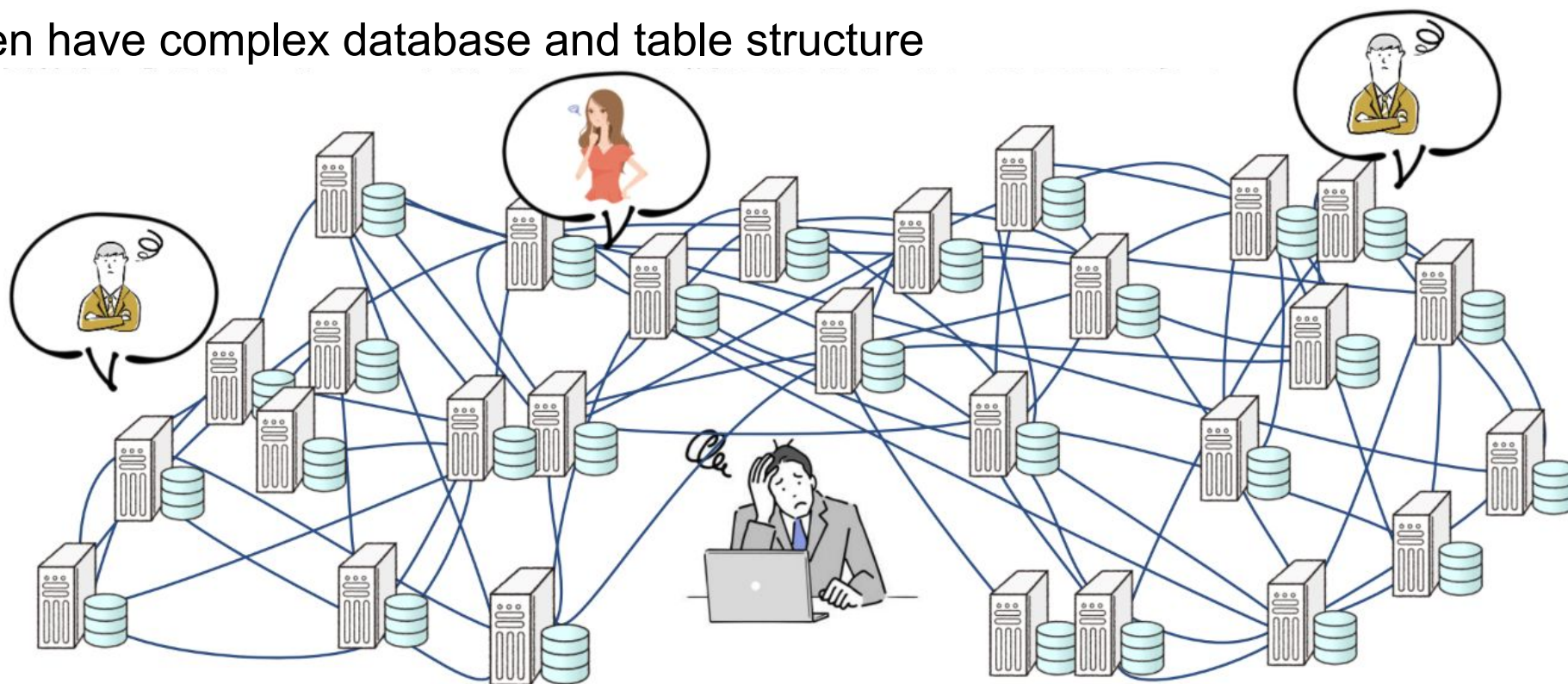
→ **Need to preprocess and store in a relational database**

- ETL = **Extract**, **Transform**, **Load**
- Facilitate smooth data analysis
- Prepare data for machine learning
- Ensure data consistency and quality across systems



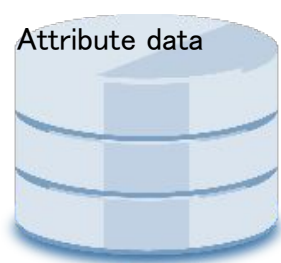
Challenges with Legacy Systems

- According to Experian, 89% of the companies face challenges with legacy systems
 - Difficulty in connecting with recently updated or modified systems
 - Often have complex database and table structure

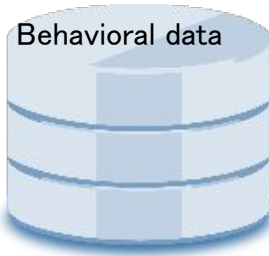


(From Week 1) It is important to use data strategically

In order to make use of data, it is essential to integrate, process, and manage various customer data (such as attribute data and behavioral data), which is why "data strategy" becomes important.



Behavioral data



??? Data



??? Data



Data can be both valuable as oil and worthless as garbage



Data should be collected *with a strategy*



Challenge

The management of old systems, their integration, additional development costs, and the expenses for integrating new systems tend to become substantial.

- Database processes data efficiently by using an optimized table design that prevents data duplication
- Relational databases organize data into collections of two-dimensional tables connected by keys
- SQL enables large-scale data processing
- As data will continue to increase, it needs to be stored in a relational database to be analyzed efficiently using SQL or Python

SQL Exercises

SQL is very similar to Pandas

SELECT

```
column_name1,  
column_name2,  
sum(column_name3)
```

**Select column,
aggregate function**

FROM

```
TABLERNAME
```

Select data

WHERE

```
column_name1 == "aaaa"
```

Specify condition

GROUP BY

```
column_name1,  
column_name2
```

Specify axis

Let's work on the notebook!

- By using databases and SQL, we can efficiently handle large volumes of data
- Learned the basics of SQL
 - CREATE and SELECT to create database/table and retrieve data
 - WHERE for conditions
 - UPDATE, DELETE, ALTER to change the data in the table
 - JOIN or LEFT JOIN to join multiple tables
 - CASE statement performs conditional logic
 - VIEW for obtaining the latest data

